

DSCI 6650: Reinforcement Learning Master of Data Science (MDSc)

Assignment-3: Reinforcement Learning - SARSA vs Q-learning in Grid World

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GitHub Repository <https://github.com/Hashna86/Project-1-for-reinforcement-learning.git>

Abstract

The goal of this assignment is to analyze and compare the performance of the SARSA and Q-learning reinforcement learning algorithms in a 5x5 Grid World environment. The task is to learn optimal navigation policies under risk of penalty and reach terminal states with minimum cumulative negative reward. Here in code implementation includes ϵ -greedy action selection - Tracks and plots reward curves - Extracts and prints learned policies - Simulates and visualizes agent trajectories for both algorithms

Environment Setup:

Grid Size is 5x5 - Start State is blue state in the position (4, 0) (bottom-left corner) - Red Penalty States are in position [(2, 0), (2, 1), (2, 3), (2, 4)] with reward -20 and Terminal States are in position [(0, 0), (0, 4)] which cause end of the episode- Other Moves: Reward -1 - Invalid Moves: Reward -1. A move that attempts to move outside of the grid receives a reward of -1.

Algorithms Implemented:

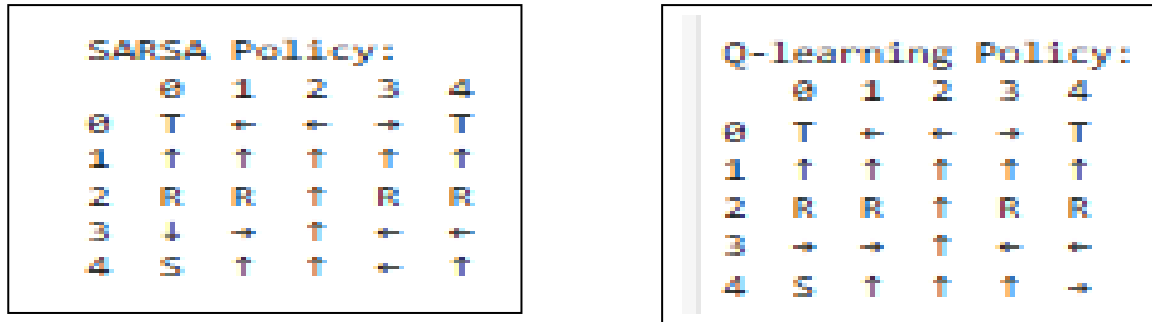
Here we implement method- 1. SARSA (On-policy TD Control) 2. Q-learning (Off-policy TD Control). Both algorithms use the ϵ -greedy policy for exploration, with the following hyperparameters: - Learning rate (α): 0.1 - Discount factor (γ): 0.95 - Exploration rate (ϵ): 0.1 - Episodes: 10,000 - Max Steps per episode: 500.

Learned Policies:

The learned policies derived from SARSA and Q-learning differ due to their underlying learning strategies. The SARSA policy is more conservative, often favoring safer and longer routes that help avoid risky regions such as red penalty states. In contrast, the Q-learning policy tends to be more direct and exploits the optimal paths that lead to

terminal states more efficiently, even if it involves moving closer to risky zones. Despite these differences, both policies successfully guide the agent to terminal states, but the paths they choose reflect the cautious nature of on-policy learning in SARSA versus the optimality-driven behavior of off-policy Q-learning

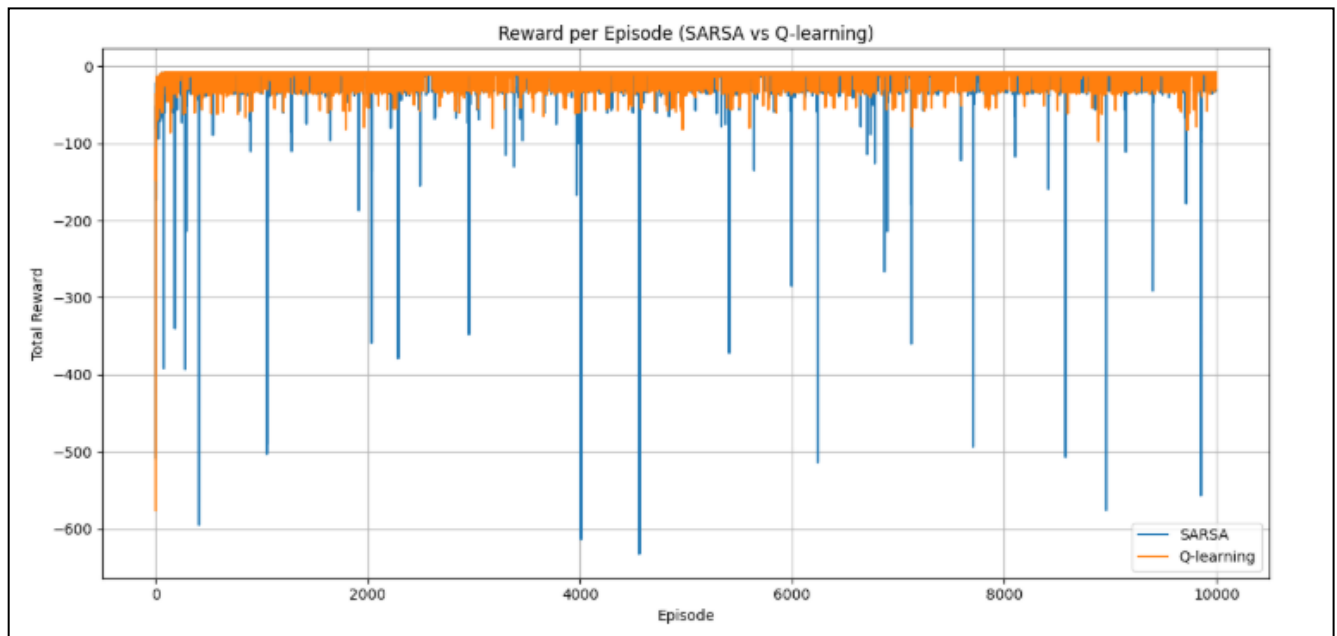
Figure-1: Visualization of SARSA Policy and Q-learning Policy



Reward curves:

The comparison of the reward trajectories over episodes shows that the SARSA and Q-learning approaches produce different results. The reward curve for Q-learning is smoother and more stable, suggesting that it learns the best course of action more quickly and consistently. The overall reward per episode in SARSA, on the other hand, frequently and significantly increases negatively. These drops result from the agent more frequently entering red penalty states during exploration, which results in severe penalties and episode resets.

Figure-2: Rewards per episode

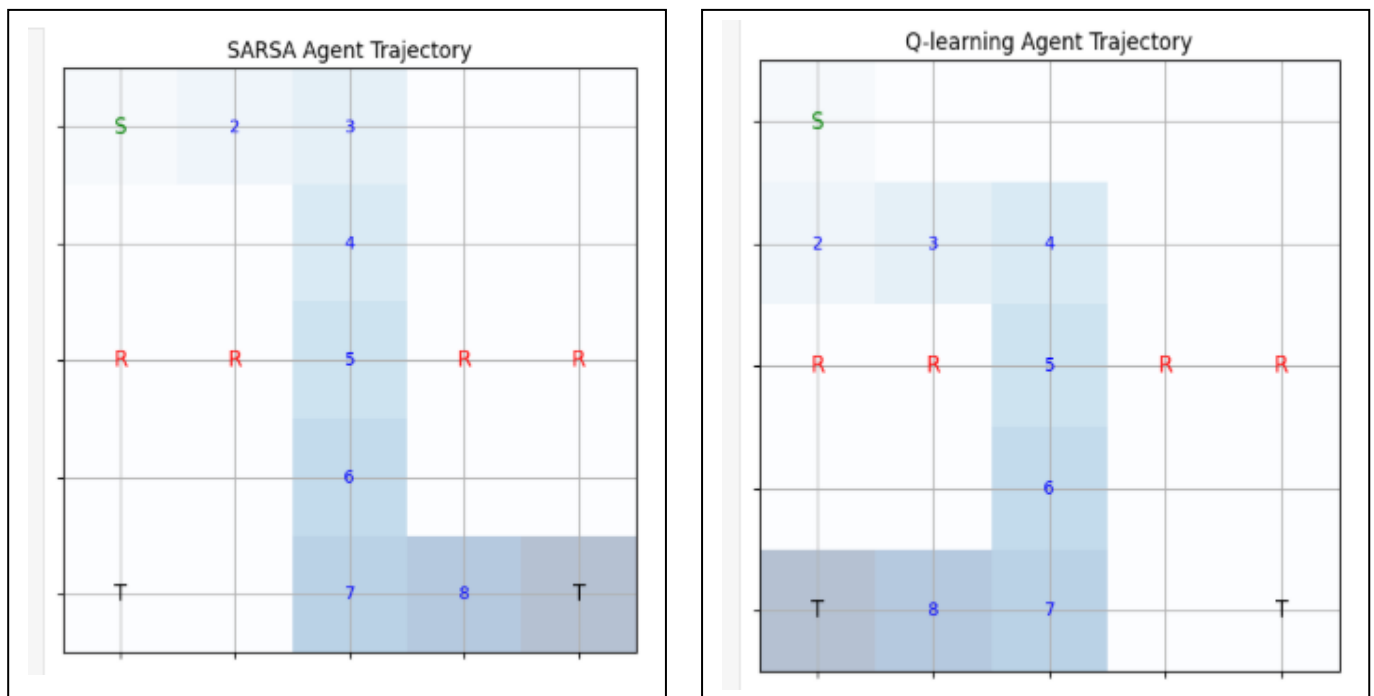


This discrepancy in performance results from each method's basic learning behavior. Because Q-learning is off-policy, it constantly adjusts its value estimations to reflect the optimal course of action going forward, resulting in quicker convergence and more effective environmental exploitation. As an on-policy approach, SARSA modifies its values in response to the actual actions performed during ϵ -greedy exploration. Because it accounts for less-than-ideal exploratory moves, it becomes more conservative but also more susceptible to volatility.

Trajectory Analysis:

SARSA Agent Trajectory: The SARSA agent follows a more cautious path, avoiding red states and reaching the terminal via a longer route. This behavior is expected due to SARSA's on-policy nature, updating based on the ϵ -greedy action it actually takes.

Figure-3: Visualization of simulated Trajectory



Q-learning Agent Trajectory: The Q-learning agent learns a more direct and optimal path by exploiting the safe gap in the red wall, even if it skirts closer to penalty zones. Its off-policy nature allows faster convergence to the optimal strategy

Comparison of Results:

Criteria	SARSA	Q-learning
Type	On-policy	Off-policy
Exploration Impact	High (affects learning path)	Low (updates toward optimal)
Trajectory Behavior	Safer, longer	Riskier, shorter
Reward Trend	Volatile	Smooth and stable
Learning Speed	Slower	Faster

Conclusion:

SARSA and Q-learning yield different behaviors and results due to their fundamental learning strategies. SARSA, being on-policy, emphasizes safer but longer routes, resulting in slower convergence and higher variance in rewards. Q-learning, as an off-policy algorithm, converges faster to the optimal policy by learning from the best possible action, leading to higher cumulative rewards and efficient paths. Overall, Q-learning performs better in this scenario both in terms of learning speed and reward consistency.